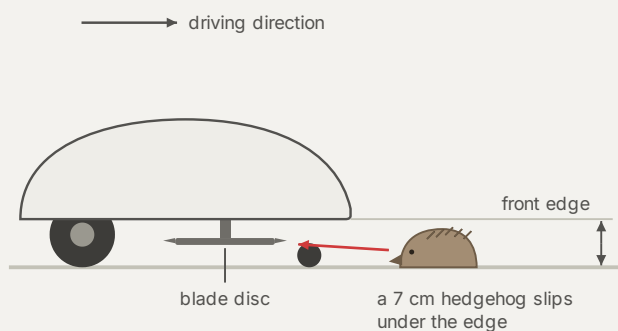


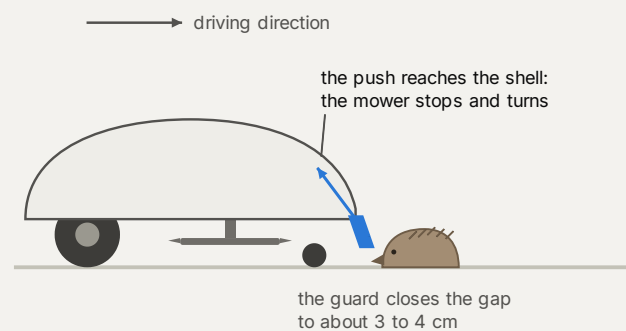
A hedgehog guard for robot mowers

The idea worked out, and the plan step by step to the final on 9 December 2026

Today: a gap under the front edge



With the guard (concept A)



Status: lead idea, not yet a team decision. Nothing in this document has been tested. The first test runs in the week of 5 October, and gate 1 decides on Sunday 11 October whether we go on with S1.

11 weeks

to the final on
Wednesday 9
December

43 steps

each with an owner, a
window and hours

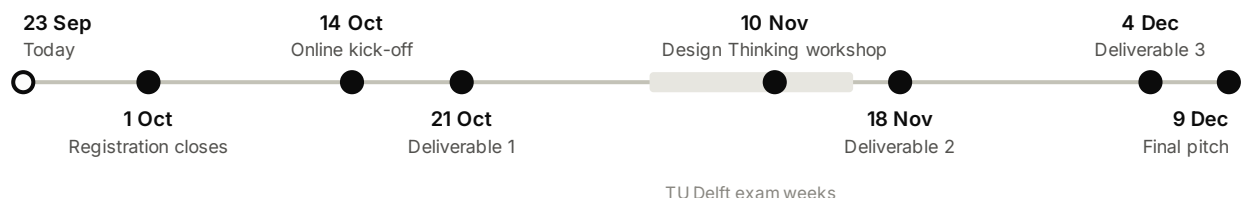
3 gates

go or no-go points on
11 Oct, 1 Nov and 29
Nov

149 h

planned, of about 183 h
the team has

The fixed dates



In one minute

- **What we build:** a clip-on guard for robot mowers that only notice obstacles by bumping into them. It makes the mower notice a young hedgehog before the blades reach it, with no wiring into the mower, and the mower still docks.
- **What we want to prove:** on the safety test that hedgehog researchers proposed in 2024, the stock mower fails with the small dummy and the guarded mower passes: only damage categories 0 to 2 (see 1.7).
- **Why it could be new:** brands are putting cameras on new models. We found no tested mechanical fix for the mowers already in gardens. That was a quick search only; the proper check is step 9.
- **What decides it first:** one comparison, in the week of 5 October: the force the mower needs to register a bump, against the force that slides a small hedgehog dummy away.
- **The time:** 11 weeks and 7 fixed dates. Deliverable 1 is due one week after the kick-off, and TU Delft exams fall in the two weeks before deliverable 2, so the heavy building happens in October.

Contents

In one minute	2
Part 1. The idea, worked out	3
1.1 The problem, in numbers	3
1.2 Who it is for	3
1.3 What we build	4
1.4 Requirements	4
1.5 The physics we have to beat	5
1.6 Four concept directions	5
1.7 How we test	6
1.8 The demo on 9 December	6
1.9 What the jury will ask	7
1.10 Tested and assumed	7
Part 2. The plan	8
2.1 The time we have	8
2.2 Timeline at a glance	8
2.3 Roles	9
2.4 Gates	9
2.5 Step by step: day by day	10
2.5 Step by step: every step in full	12
2.6 Hours per week	16
2.7 Money	17
2.8 Risks	17
2.9 Open questions	18
After the final	18
Sources	18

Part 1. The idea, worked out

1.1 The problem, in numbers

The numbers the pitch will use. Every line has a source; the links are at the end.

Claim	Source
Aalborg tested 18 robot mowers with dead hedgehogs, each with a 3 m run-up. None detected the dependent juveniles under 200 g; these often passed under the blades without visible injury	Rasmussen et al. 2021
In a follow-up with 19 models, apart from one single case no mower detected a hedgehog without touching it. Height was the size measure that most influenced injury: larger hedgehogs were injured less	Rasmussen et al. 2024
Skid plates, pivoting blades and front-wheel drive were linked to less damage	Rasmussen et al. 2021
The Husqvarna mowers in that test drove at 21 to 39 m per minute, which is 0.35 to 0.65 m/s	Rasmussen et al. 2021
In Germany, 370 hedgehogs injured by garden tools were recorded from June 2022 to September 2023, and 47% died	Leibniz-IZW
The European hedgehog has been Near Threatened on the IUCN Red List since 29 October 2024	Hedgehog Street
Wallonia bans robot mowers from 18:00 to 9:00 since 9 April 2026. The Netherlands has no such rule	Walloon environment department; NOS
Young hedgehogs are born in late summer and early autumn. Hibernation usually runs from October or November to March or April	Egelbescherming Nederland

Our reading of the two studies, not a finding of the papers: the smallest hedgehogs can slip under the blades, the big ones get bumped, and the ones in between get hit. The small dummy in the proposed test (7 cm high) sits in that band.

Not established, so never said as fact: how many hedgehogs mowers injure in the Netherlands each year, and how many contact-only mowers run in Dutch gardens (see 1.10).

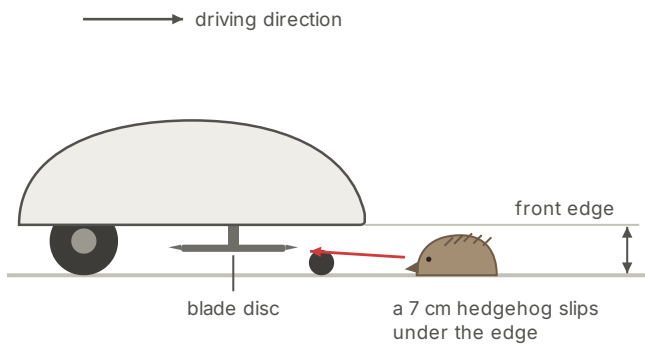
1.2 Who it is for

- **Owners of contact-only mowers already in their garden:** they want a mower that works and no dead hedgehogs, without buying a new mower. Whether they would fit a guard, and at what price, is an **assumption** (owner survey, steps 23 and 25).
- **Hedgehog organisations** (Egelbescherming Nederland, local shelters): today they advise mowing only by day. A tested guard would give them a second message. Whether they would back it is an **assumption** (step 4).
- **Researchers and mower makers** (Aalborg and Oxford; IZW and the crash-test firm CTS in Germany, who are working towards a standard): our test data are useful to them even if nobody ever sells our guard.

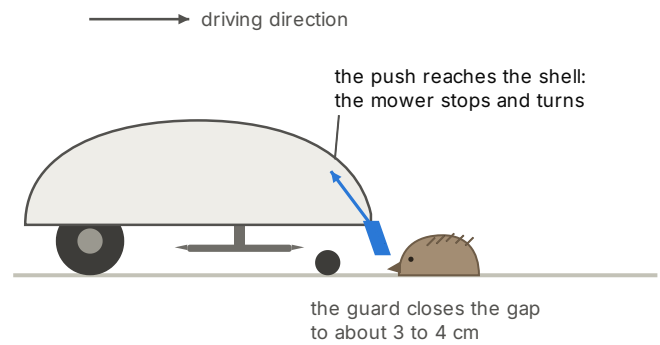
1.3 What we build

A light guard that clips onto the mower's body shell at the front and front corners. On contact-only mowers the shell floats on the chassis, and moving it trips the collision sensor (deep dive; to confirm for our model in step 6). The guard closes the gap under the front edge, where a 7 cm hedgehog now slips under, and turns a small push low down into a bump the mower already knows how to handle: stop, reverse, turn.

Today: a gap under the front edge



With the guard (concept A)



Schematic, not to scale. The heights are placeholders until we measure our own mower in steps 6 and 13. The red arrow is the hedgehog's path relative to the moving mower.

First platform: a used Husqvarna Automower 105. It is a common entry model. One Marktplaats search on 23 September showed used ones asking €75 to €500, most between €100 and €300. A common model also means many possible users and cheap spare parts.

1.4 Requirements

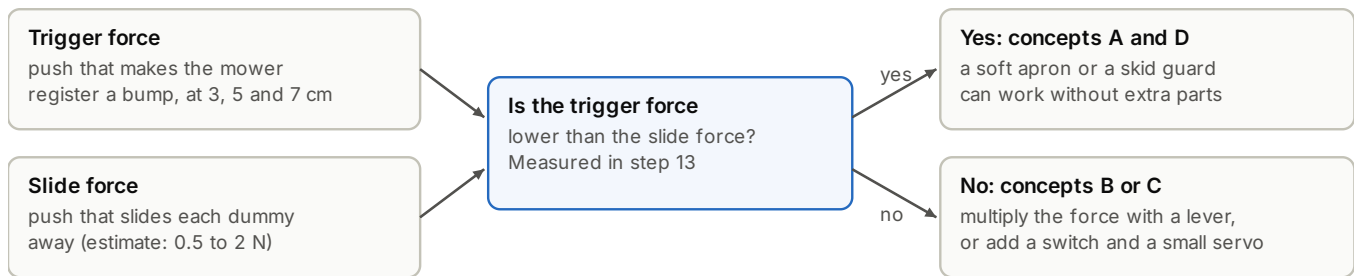
#	Requirement	Target	Where the number comes from
R1	Small dummy: 7 cm high, under 400 g	Damage categories 0 to 2 in 60 of 60 trials	The 2024 protocol and its labelling rule
R2	Large dummy: at least 10 cm high, over 600 g	Categories 0 to 2 in 60 of 60 trials	Same
R3	Do no harm to the smallest: a tiny dummy of about 5 cm and 180 g	No more category 3 or 4 than the stock mower	Our own addition. The 5 cm is an assumption until a shelter confirms it
R4	Still docks	10 of 10 dockings	A German sheet-metal guard on an Automower 450X failed on exactly this (robomaeher.de)
R5	No extra stops on real grass	No more stops per 30 minutes than the stock mower	Our target
R6	Fits without tools or wiring, and comes off in a minute	Yes or no	Our target: a retrofit has to be easy
R7	Works at the mower's own speed	Tested at stock speed	0.35 to 0.65 m/s for Husqvarna models (Rasmussen 2021)
R8	Material cost	Under €15	Assumption , our guess at what owners would accept; the survey tests it

1.5 The physics we have to beat

Two numbers decide the concept. We measure both in the week of 5 October (step 13):

1. **Trigger force:** how hard you must push the shell, at 3, 5 and 7 cm above the ground, before the mower registers a bump. A luggage scale or a kitchen scale is enough.
2. **Slide force:** how hard you must push each dummy on the test surface before it slides. Our estimate, not a measurement: with a friction coefficient of 0.3 to 0.6, about 0.5 to 1 N for 180 g and 1 to 2 N for 350 g.

If the trigger force is higher than the slide force, a stiff guard shoves the dummy ahead until the mower rolls over it (category 3 or 4). The guard then has to multiply force (concept B) or bring its own trigger (concept C).



A third number sets how far ahead the guard must reach. At 0.35 to 0.65 m/s the mower travels 3.5 to 6.5 cm for every 0.1 s it takes to react. We film the stop in slow motion, measure the reaction time and the stopping distance, and make the guard touch the dummy at least that far ahead of the blade disc.

1.6 Four concept directions

A. Low apron

How it works

A flexible skirt (TPU flaps or a brush strip) lowers the front edge to about 3 to 4 cm and passes a push into the shell

Why it might work

No electronics, cheapest to make

Main risk

Needs more force than the dummy resists, so it shoves it; drags on grass

First test

Cardboard or PETG version, one afternoon

B. Lever feeler

How it works

A light bar low down, hinged on the shell, with a lever that pushes against the chassis, so a small touch moves the shell enough to trip the sensor

Why it might work

Pure mechanism, and it solves the force problem

Main risk

Needs a solid point on the chassis to push against; travel and play

First test

Wood-and-hinge mock-up once we know the trigger force

C. Switch and kick

How it works

A low flap with a microswitch; a small battery servo nudges the shell, so the mower's own logic reacts

Why it might work

The trigger force can be as low as we like

Main risk

More parts, a battery, rain, reaction time

First test

Arduino, microswitch and servo on the bench

D. Skid guard

How it works

A low guard around the blade disc that keeps a hedgehog-sized body out of blade reach even when detection fails

Why it might work

Skid plates were linked to less damage (Rasmussen 2021)

Main risk

Docking, which is where the German guard failed; cut quality

First test

Cardboard ring; test docking first

Order: A and D first, one afternoon each (step 14). B or C only if step 13 shows that A cannot trip the sensor below the slide force. The answer may be a combination, such as A with D.

1.7 How we test: our version of the published protocol

From Rasmussen et al. 2024, with our changes marked **ours**.

- **Surface:** a coconut mat with rubber backing on a concrete floor, 2 × 5 m, 20 mm thick. **Ours:** the same kind of mat but 1 m wide, because the mower drives straight and only the dummy turns; or a lawn for development runs. Stock and guarded runs always use the same surface.
- **Placement:** the dummy 1 m from the end of the mat, the mower 3 m away.
- **Positions:** the dummy standing, snout at 12 o'clock (facing the mower), at 2 to 3 o'clock and at 9 to 10 o'clock. 20 trials each, so 60 per dummy.
- **Cameras:** two, 1 m from the dummy, one at the side and one behind. **Ours:** two phones filming in slow motion.
- **Cutting height:** fixed and written down. The 2021 study used the highest setting.
- **Dummies:** the paper gives size classes but no build ("work is currently underway to design an optimal hedgehog crash test dummy"). **Ours:** a foam or printed core, sand for weight, and a clay or paint skin so blade contact leaves marks. Weigh, measure and photograph each dummy: the jury will ask.
- **Development runs:** 5 per position, 15 per dummy, to learn fast. **Final runs:** the full 60 per dummy.

How each run is scored

 0 Senses the hedgehog from a distance and drives on without touching it	 1 Touches it lightly (a nudge) and thereby detects it	 2 Touches it (a flip), moving it into a different position, and detects it	 3 Fails to detect it and drives across it	 4 Drives across it, and the blades cause injuries
--	--	---	--	--

Counts as safe for hedgehogs, if every trial lands here

Cannot be labelled safe

Fails the test

Categories as defined in the 2021 study; labelling rule from the 2024 paper. We still check that Table 2 of the 2024 paper uses the same definitions.

Run log, one row per run, as a CSV next to the videos:

```
date,run,mower,guard_version,surface,dummy,position,category,pushed_cm,docked,notes,video_file
```

Safety: one person operates and keeps a hand near the stop button; nobody within reach while the mower runs; gloves when touching blades; blades off for the stage, and we say so. **Dummies only, never live animals.**

1.8 The demo on 9 December

- **Live:** the guarded mower drives at the small dummy on our mat and backs off. Blades off on stage.
- **Proof:** a slow-motion clip, stock and guarded side by side, same dummy, same spot.
- **In the jury's hands:** a spare guard to press with one finger, to feel how little it takes.
- **Our own rule** (CLAUDE.md): the demo works without Wi-Fi and without the presenter's laptop, and we rehearse once on a borrowed device. The clip also sits on a USB stick and on two phones.
- **Ask the organiser in the week of 16 November** (step 34): floor space of about 1 × 5 m, a power socket for the charging station, and whether we can lay the boundary wire. Wired mowers usually will not drive without the wire signal (check the 105 manual). If the stage cannot take a running mower, the clip carries the demo.

Pitch line, draft: "Scientists tested 19 robot mowers. Apart from one single case, none noticed a hedgehog before bumping into it. We made an ordinary mower back off in time, and tested it on the scientists' own protocol: [our result]." We only say "passes" once the 120 final runs say so.

1.9 What the jury will ask, and our answer

Question	Answer, and what backs it
Why not just mow by day?	That is the best advice and we say so (Egelbescherming Nederland). But there is no Dutch rule (NOS), the 2021 authors note that hedgehogs are sometimes active by day, and young ones stay active into autumn (Egelbescherming). A guard protects whatever the schedule
New mowers have cameras now	New models, yes: Husqvarna's AI vision for 2026, and Navimow's own claim of 95% avoidance. The mowers already in gardens do not, and they will run for years. Assumption: most mowers in Dutch gardens are contact-only (step 10)
Does it fit my mower?	Our tests use the Automower 105. One fit check on a second model if time allows; otherwise we say plainly that one model was tested
Where do your numbers come from?	Every problem number has a source. Every result comes from our run log and videos
Who would make and sell it?	Options, none tested: printable files shared through hedgehog groups, an accessory sold by dealers, or our data handed to makers and to the standard work. A retrofit that changes how a certified machine senses collisions raises warranty and product-safety questions, so we ask a dealer and our Samsung mentor which route is realistic

1.10 Tested and assumed, kept apart

Tested by us: **nothing yet**. Sourced: everything in 1.1. The assumptions, and the step that tests each:

Assumption	Tested in
Our stock mower fails with the small dummy (categories 3 and 4)	Step 12
A guard can trip the sensor below the dummy's slide force	Steps 13 and 14
It can do that and still dock, without extra stops on grass	Steps 14, 24 and 37
Owners would fit a guard, and under €15 of material is the right target	Steps 23, 25 and 30
Hedgehog organisations would back a retrofit	Step 4
Most mowers in Dutch gardens are contact-only	Step 10
A dependent juvenile is about 5 cm high	Step 4 (the shelter)

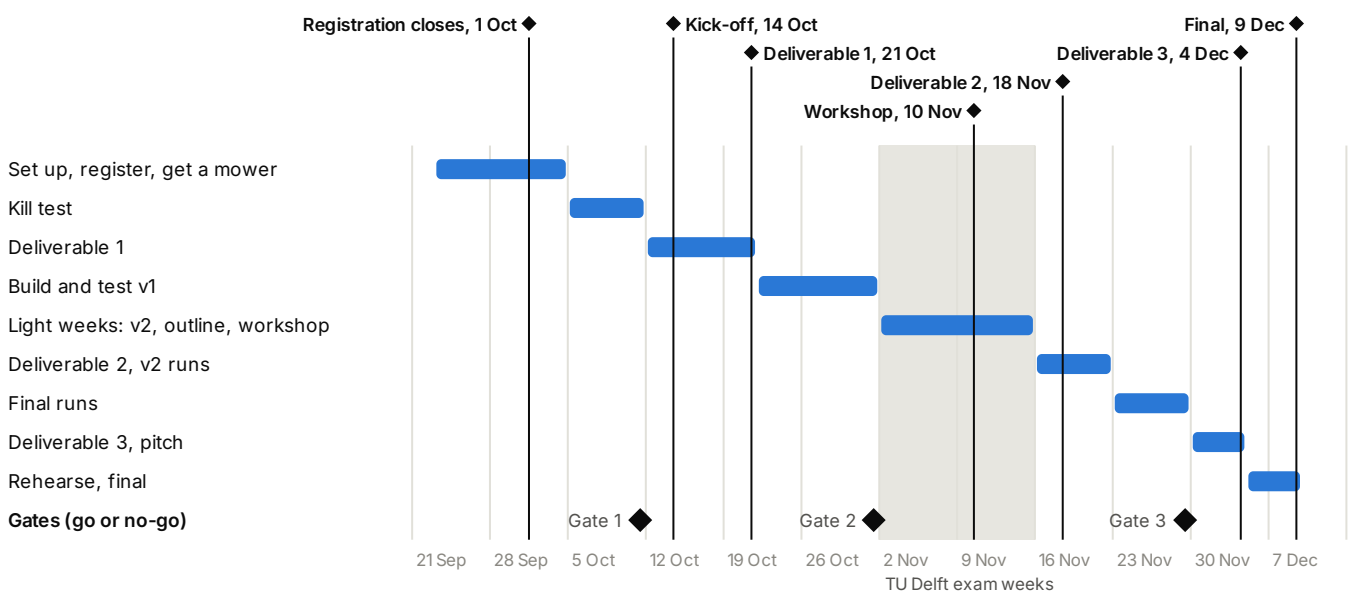
Part 2. The plan

2.1 The time we have

- **77 days, exactly 11 weeks**, from today, Wednesday 23 September, to the final on Wednesday 9 December.
- **The tight spots:** deliverable 1 is due one week after the kick-off, deliverable 3 sixteen days after deliverable 2, and the final five days after deliverable 3.
- **Teams may be cut** ("the best teams progress to the next round", from the brief), and we do not know after which deliverable. So deliverable 1 must show a test result, not only an idea. That is why the kill test sits before the kick-off.
- **Exams:** in the TU Delft calendar, 2 to 8 November is limited teaching or exams (varies per programme) and 9 to 15 November is exams. The Design Thinking workshop on Tuesday 10 November falls in that exam week. If you are not on the TU Delft calendar, put your own exam weeks in the table below.
- **Hours:** the plan assumes 6 hours a week each, 18 for the team, and half that in the two exam weeks. That gives about 183 hours of capacity against 149 planned, a buffer of about 34 hours. Everyone replaces the 6 with their real number at the first meeting.

Name	Hours a week	Busy weeks (exams, work, trips)
Hoang		
Teammate 2		
Teammate 3		

2.2 Timeline at a glance



Blue bars are our phases, black diamonds and lines are the competition dates and our three gates, gray is the TU Delft exam period.

2.3 Roles

Three leads, so every step has one name on it. Everyone helps everywhere; the lead owns the deadline. Names go in at the first meeting. The Story lead has the lightest load, so the Story lead is usually the "+ 1" who helps on test days.

■ Build lead

about 46 h

Concepts, CAD, printing, fitting the guard

■ Test lead

about 48 h

The mower, dummies, test lane, runs, videos and run log

■ Story lead

about 37 h

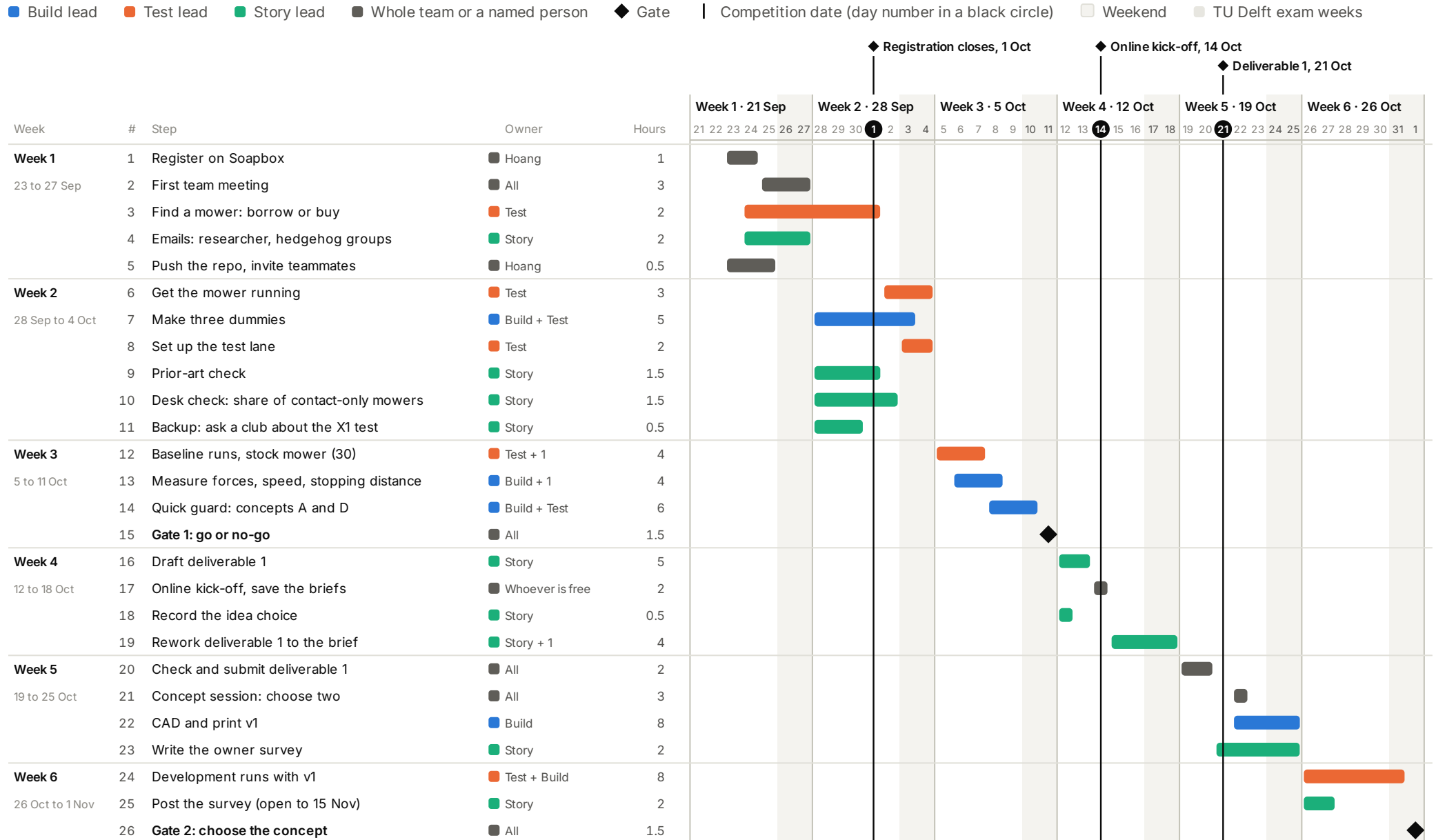
Registration, sources, stakeholders, survey, deliverables and pitch

2.4 Gates

Gate	When	Go on if	If not
1. S1 go or no-go	Sunday 11 October	The stock mower scores 3 or 4 in some small-dummy runs, and either the quick guard turns some of those into 0 to 2 while still docking, or the numbers from step 13 show that a lever (concept B) can get below the slide force	If our mower already detects the small dummy every time, try a second model before dropping S1. Otherwise switch to X1: its boot test runs in the week of 12 October (step 11) and deliverable 1 is written on X1
2. Concept choice	Sunday 1 November	One concept has the fewest 3 and 4 results in the development runs, docks 5 of 5, and adds no stops in 15 minutes on grass	Build concept B or C over the exam weeks, and report the negative result honestly in deliverable 2
3. Design freeze	Sunday 29 November	The final runs are done	We pitch the numbers we have. After this date: fixes only, no new features

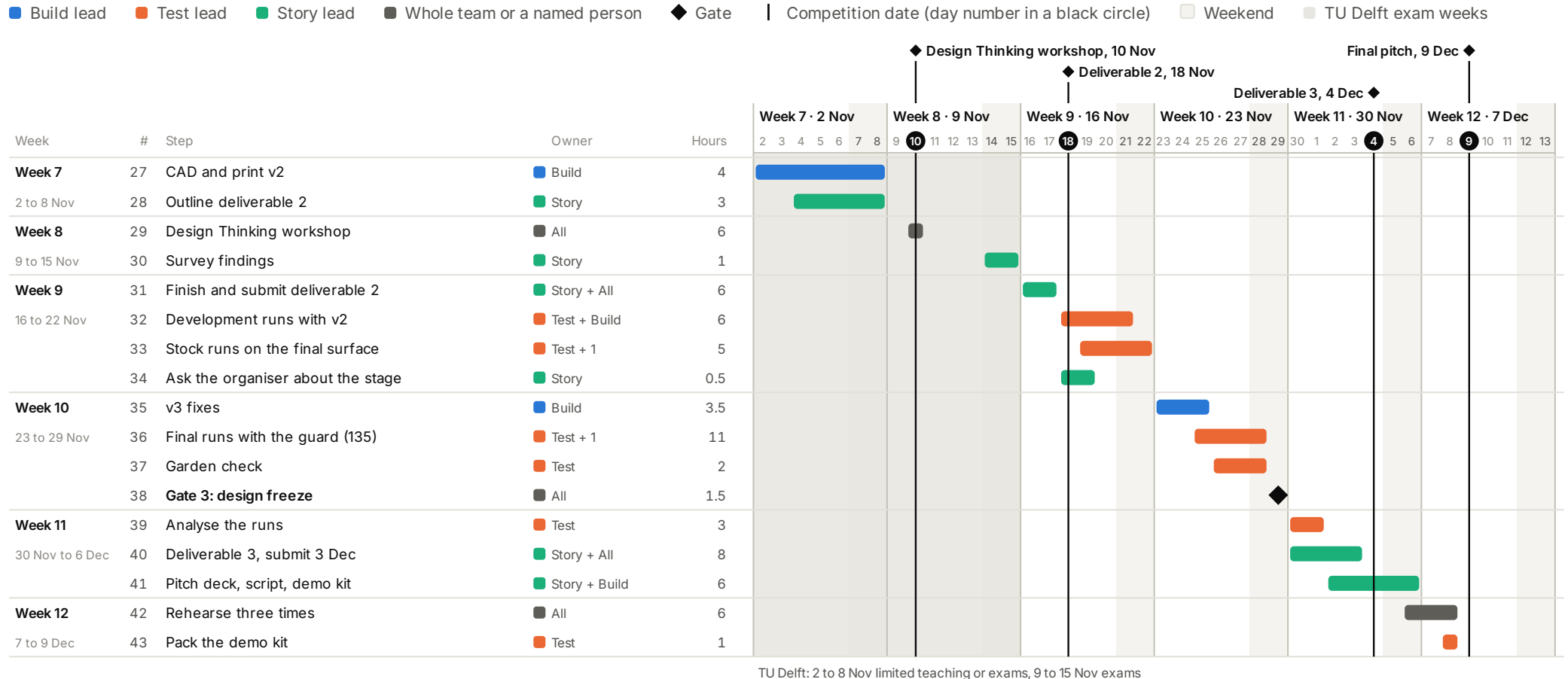
2.5 Step by step: day by day, weeks 1 to 6

Each bar is one step, placed in its planned window. Colour shows the lead who owns the deadline; with two names, the first one owns it. The full text of every step, with its "done when", follows after these two pages.



2.5 Step by step: day by day, weeks 7 to 12

Same scale as the previous page. Weeks 9 to 11 run at or near capacity: when an earlier week runs light, pull work forward.



2.5 Step by step: every step in full

The owner is a role from 2.3. "+ 1" means one more person, usually the Story lead. "When" is the planned window, not a hard deadline, except where it names a submission. Hours are team hours and are our estimates.

Week 1 Wednesday 23 to Sunday 27 September

8.5 h planned of 13

Set up

#	Step	Owner	When	Hours	Done when
1	Register the team on Soapbox now. The page lists 100 spots, so do not wait for 1 October. Have both teammates' names and emails at hand. Save what the form asks, word for word, in <code>Resources/</code>	■ Hoang	Wed 23 to Thu 24 Sep	1	Confirmation received
2	First team meeting, one hour: agree S1 as the lead idea to test, pick the three roles, fill in the hours table, add names and strengths to <code>MEMORY.md</code> , collect GitHub usernames, fix a weekly 30-minute check-in (Sunday evening lines up with the gates)	■ All	Fri 25 to Sun 27 Sep	3	Roles and hours filled in
3	Find a mower. Ask family, friends and neighbours, and shortlist three used contact-only mowers that come with charging station and wire (Automower 105 or similar). Decide: borrow or buy	■ Test	Thu 24 Sep to Thu 1 Oct	2	A mower promised for pickup by Sunday 4 October
4	Send three emails: Dr Sophie Lund Rasmussen (how she builds dummies, would she look at our data), Egelbescherming Nederland (would they comment on a retrofit), and a hedgehog shelter near Den Haag or Delft (how tall juveniles are by weight, do they see mower injuries)	■ Story	Thu 24 to Sun 27 Sep	2	Sent, reminder set for 7 days later
5	Push the repo and invite both teammates (open since 23 September)	■ Hoang	Wed 23 to Fri 25 Sep	0.5	Both can pull

Week 2 Monday 28 September to Sunday 4 October

13.5 h planned of 18

Registration closes Thursday 1 October

#	Step	Owner	When	Hours	Done when
6	Get the mower running. Collect it, charge it, and read the manual on collision and lift sensing, the boundary wire, cutting height and taking the blades off	■ Test	Fri 2 to Sun 4 Oct	3	It drives, and stops when it bumps into something
7	Make three dummies: small (7 cm high, 300 to 400 g), large (at least 10 cm, 600 to 800 g), tiny (about 5 cm and 180 g, an assumption until the shelter answers). A skin that shows blade marks. Weigh, measure, photograph	■ Build + Test	Mon 28 Sep to Sat 3 Oct	5	Three dummies logged with photos
8	Set up the test lane: lawn or mat, boundary wire around it, start line at 3 m, a mark for the dummy, two phone stands	■ Test	Sat 3 to Sun 4 Oct	2	Five practice runs work
9	Prior-art check: Espacenet (robotic mower with hedgehog or animal, and "Mähroboter Igel"), Printables, Thingiverse, German mower forums	■ Story	Mon 28 Sep to Thu 1 Oct	1.5	Findings in <code>MEMORY.md</code> . A tested guard that already exists goes to gate 1
10	Desk check: what share of robot mowers in Dutch gardens only detect by contact, and how many there are	■ Story	Mon 28 Sep to Fri 2 Oct	1.5	A sourced number, or "not found"
11	Backup for gate 1, optional: ask one football or hockey club if we can brush boots after a training in the week of 12 October (the X1 test)	■ Story	Mon 28 to Wed 30 Sep	0.5	A yes or no from a club

Week 3 Monday 5 to Sunday 11 October

15.5 h planned of 18

Kill test. TU Delft week 1.5 can hold exams for some programmes: check the busy-weeks table

#	Step	Owner	When	Hours	Done when
12	Baseline runs: stock mower, small and large dummy, 5 runs in each of the 3 positions, so 30 runs, each scored 0 to 4	Test + 1	Mon 5 to Wed 7 Oct	4	30 rows in the run log
13	Measure: trigger force at 3, 5 and 7 cm; slide force of each dummy; speed over 3 m; reaction time and stopping distance from slow motion	Build + 1	Tue 6 to Thu 8 Oct	4	Numbers in the log
14	Quick guard: cardboard or PETG versions of concepts A and D. Repeat the 30 runs with the better one, dock 5 times, mow 10 minutes of real grass	Build + Test	Thu 8 to Sat 10 Oct	6	Results in the log
15	Gate 1 meeting, Sunday 11 October: go or no-go, written in MEMORY.md	All	Sun 11 Oct	1.5	Decision recorded

Week 4 Monday 12 to Sunday 18 October

11.5 h planned of 18

Online kick-off Wednesday 14 October

#	Step	Owner	When	Hours	Done when
16	Draft deliverable 1 before the kick-off in Deliverables/1 - <name>/: the problem with sources, the idea, the first test result, the plan	Story	Mon 12 to Tue 13 Oct	5	Draft ready Tuesday 13 October
17	Kick-off, Wednesday 14 October, online. Hoang has another commitment that evening, so a teammate joins. Save every brief and the judging criteria word for word in Resources/ that night	Whoever is free	Wed 14 Oct	2	Briefs in Resources/
18	Record the idea choice in MEMORY.md under Key Decisions: date, who, why	Story	Mon 12 Oct	0.5	Recorded
19	Rework the deliverable 1 draft to the real brief	Story + 1	Thu 15 to Sun 18 Oct	4	Draft covers every point of the brief

Week 5 Monday 19 to Sunday 25 October

15 h planned of 18

Deliverable 1 due Wednesday 21 October

#	Step	Owner	When	Hours	Done when
20	Check deliverable 1 line by line against the brief. Submit on Tuesday 20 October, a day early	All	Mon 19 to Tue 20 Oct	2	Submitted
21	Concept session: sketch A to D with the measured forces, choose two to build	All	Thu 22 Oct	3	Two concepts chosen, reasons in MEMORY.md
22	CAD and print v1 of both concepts	Build	Thu 22 to Sun 25 Oct	8	Both fit the mower
23	Owner survey: 8 questions (which mower, when it mows, would you fit a guard, what would you pay, where would you buy it)	Story	Wed 21 to Sun 25 Oct	2	Survey ready to post

Week 6 Monday 26 October to Sunday 1 November

11.5 h planned of 18

Build and test v1

#	Step	Owner	When	Hours	Done when
24	Development runs with both v1 concepts: 15 per dummy each, 5 dockings, 15 minutes on grass	Test + Build	Mon 26 to Sat 31 Oct	8	Results in the log

#	Step	Owner	When	Hours	Done when
25	Post the survey in hedgehog and garden groups and neighbourhood apps. Target: 30 answers by 15 November	Story	Mon 26 to Tue 27 Oct	2	Posted
26	Gate 2 meeting, Sunday 1 November: choose the concept	All	Sun 1 Nov	1.5	Choice in MEMORY.md

Week 7 Monday 2 to Sunday 8 November

7 h planned of 9

Exams for some of us: a light week

#	Step	Owner	When	Hours	Done when
27	CAD and print v2 of the chosen concept	Build	Mon 2 to Sun 8 Nov	4	Printed
28	Outline deliverable 2 with what we have: problem, user insight, concept, first data	Story	Wed 4 to Sun 8 Nov	3	Outline in Deliverables/2 - <name>/

Week 8 Monday 9 to Sunday 15 November

7 h planned of 9

Exams; Design Thinking workshop Tuesday 10 November

#	Step	Owner	When	Hours	Done when
29	Design Thinking workshop, Tuesday 10 November. At least one of us, all three if exams allow. Notes and material into Resources/	All	Tue 10 Nov	6	Notes saved
30	Read the survey answers: three findings for deliverable 2	Story	Sat 14 to Sun 15 Nov	1	Findings in the draft

Week 9 Monday 16 to Sunday 22 November

17.5 h planned of 18

Deliverable 2 due Wednesday 18 November

#	Step	Owner	When	Hours	Done when
31	Finish deliverable 2 , check it against the brief, submit on Tuesday 17 November	Story + All	Mon 16 to Tue 17 Nov	6	Submitted
32	v2 development runs: 15 per dummy, 5 dockings	Test + Build	Wed 18 to Sat 21 Nov	6	Results in the log
33	Stock runs on the final surface: small dummy up to 60 runs (all 60 again if the surface changed since week 3), tiny dummy 15 runs	Test + 1	Thu 19 to Sun 22 Nov	5	Stock baseline complete
34	Ask the organiser about the final: floor space, power socket, set-up time, whether a mower may drive on stage	Story	Wed 18 to Thu 19 Nov	0.5	Answer saved in Resources/

Week 10 Monday 23 to Sunday 29 November

18 h planned of 18

Final runs

#	Step	Owner	When	Hours	Done when
35	v3 fixes: fit, docking, grass	Build	Mon 23 to Wed 25 Nov	3.5	Docks 10 of 10
36	Final runs with the guard: 60 with the small dummy, 60 with the large, 15 with the tiny one	Test + 1	Wed 25 to Sat 28 Nov	11	Full run log and videos
37	Garden check: three times 30 minutes of mowing with the guard, filmed, stops counted and compared with the stock mower	Test	Thu 26 to Sat 28 Nov	2	Stops per 30 minutes, stock and guard

#	Step	Owner	When	Hours	Done when
38	Gate 3, Sunday 29 November: design freeze	■ All	Sun 29 Nov	1.5	Numbers final

Week 11 Monday 30 November to Sunday 6 December 17 h planned of 18

Deliverable 3 due Friday 4 December

#	Step	Owner	When	Hours	Done when
39	Analyse the runs: three numbers and one chart (damage categories, stock against guard, per dummy)	■ Test	Mon 30 Nov to Tue 1 Dec	3	Checked by a second person
40	Deliverable 3: draft Monday to Wednesday, check it against the brief, submit Thursday 3 December	■ Story + All	Mon 30 Nov to Thu 3 Dec	8	Submitted
41	Pitch deck, script and demo kit: side-by-side clip, a guard to press, mat, dummies, mower with blades off	■ Story + Build	Wed 2 to Sun 6 Dec	6	First full run-through

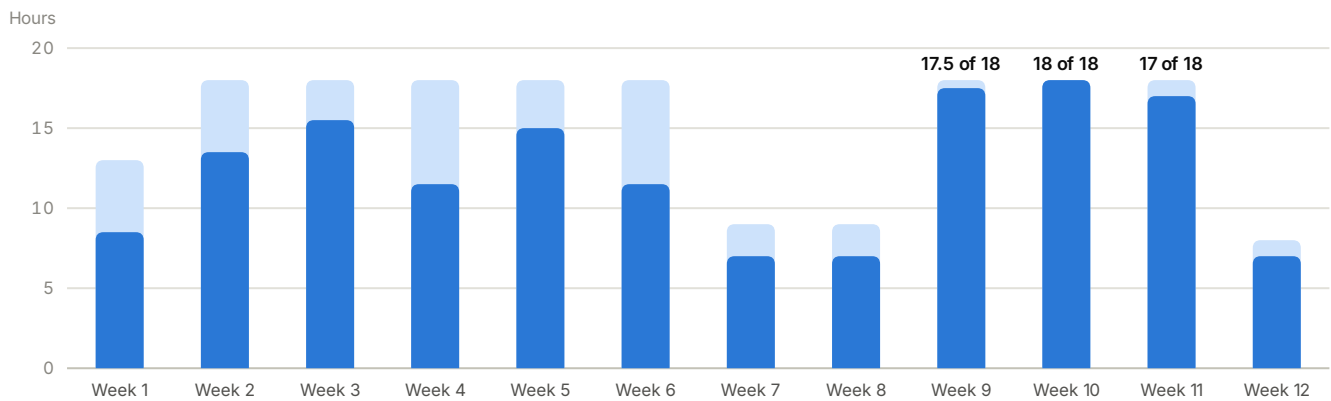
Week 12 Monday 7 to Wednesday 9 December 7 h planned of 8

Final pitch Wednesday 9 December

#	Step	Owner	When	Hours	Done when
42	Rehearse three times , timed: once on a borrowed device, once with Wi-Fi off	■ All	Sun 6 to Tue 8 Dec	6	Within the time limit, demo works offline
43	Pack the demo kit the evening before. Clip on a USB stick and on two phones	■ Test	Tue 8 Dec	1	Checklist ticked

2.6 Hours per week

■ Planned hours ■ Capacity at 6 hours each (half in exam weeks, part-weeks prorated)



Week 1 and week 12 are part-weeks, and weeks 7 and 8 count at half for the exams. Weeks 9 to 11 are full, so when an earlier week runs light, pull work forward: the pitch outline, the analysis sheet, the demo kit list.

Week	Dates	Planned hours	Capacity
Week 1	Wednesday 23 to Sunday 27 September	8.5	13
Week 2	Monday 28 September to Sunday 4 October	13.5	18
Week 3	Monday 5 to Sunday 11 October	15.5	18
Week 4	Monday 12 to Sunday 18 October	11.5	18
Week 5	Monday 19 to Sunday 25 October	15	18
Week 6	Monday 26 October to Sunday 1 November	11.5	18
Week 7	Monday 2 to Sunday 8 November	7	9
Week 8	Monday 9 to Sunday 15 November	7	9
Week 9	Monday 16 to Sunday 22 November	17.5	18
Week 10	Monday 23 to Sunday 29 November	18	18
Week 11	Monday 30 November to Sunday 6 December	17	18
Week 12	Monday 7 to Wednesday 9 December	7	8
Total		149	183

2.7 Money

Item	Cost	Source
Used Automower 105 with charging station and wire	€75 to €500 asking, most €100 to €300	Marktplaats search, 23 September 2026
Or borrow one	€0, but we cannot modify it freely or keep it until the final	Our judgement
Coconut mat 1 × 5 m, optional, for the final runs	About €160	From €32.50 per m ² for a 17 mm roll (rubbermatten24.nl); the protocol mat is 20 mm with rubber backing, so check the exact product
PETG and TPU filament	About €40	Estimate
Dummy materials: foam, sand, clay, paint	About €30	Estimate
Luggage scale for the force measurements	About €10	Estimate

Without the mat, that is about €155 to €380 with a used mower at €75 to €300, or about €80 if we borrow one. Split three ways, about €25 to €130 each. Selling the mower after the final gets part of it back (**assumption**).

2.8 Risks

Risk	Early signal	What we do
No mower by Sunday 4 October	No yes from step 3 by Thursday 1 October	Buy a used one; do not wait for a loan
The shell needs more force than a small dummy resists	Step 13	Concept B or C. That is why we measure first
The guard stops the mower docking	The first docking test in step 14	Test docking from the cardboard version on; shape the guard around the charging station
False stops on grass	Steps 24 and 37	Softer flaps or brushes; compare with the stock mower's stop rate
Someone already sells a tested guard	Step 9	Our claim becomes the protocol test itself; decide at gate 1
The deliverable briefs ask for something we did not plan	The kick-off, step 17	Re-plan the week after the kick-off. Until then this plan is a draft
Exams in weeks 7 and 8, and for some programmes in week 3	The busy-weeks table	Heavy building in weeks 5 and 6; weeks 7 and 8 stay light; move the kill test a few days if week 3 clashes
Mowers go into storage and hedgehogs hibernate from October or November	Now	Test with dummies on a mat or lawn; a garden field trial becomes the next step after the final
Stakeholders do not answer	7 days after step 4	One reminder, then a phone call. The plan does not depend on them
The stage cannot take a running mower	Step 34	The side-by-side clip carries the demo
Someone gets cut by the blades	Always	One operator, stop button, nobody within reach, gloves, blades off on stage
A retrofit on a certified machine raises warranty or safety questions	A jury question	Say it openly; ask a dealer and our Samsung mentor what a realistic route is

2.9 Open questions

Question	Cheapest way to close it	By
What registration asks, and whether it needs the idea	Step 1	Sunday 27 September
What each deliverable must contain, the judging criteria, the language	The kick-off (step 17); if not answered there, ask through Soapbox	Wednesday 14 October
After which deliverable teams are cut	The kick-off	Wednesday 14 October
The kick-off time	The registration confirmation or the Soapbox page	Thursday 1 October
The final venue, floor space and power	Step 34	Sunday 22 November

After the final

- Whatever happens on 9 December, the results and lessons go into `MEMORY.md`.
- The real test is a field trial in gardens, and that can only start when mowers and hedgehogs are out again, around March or April 2027 (Egelbescherming). If we win, that is what the €2,500 would pay for: guards for a handful of gardens, the proper coconut mat, and dummies built to the protocol. This is our proposal and is not costed yet.

Sources

Evidence

[Rasmussen et al. 2021, Animals: robot mower tests with hedgehogs](https://www.mdpi.com/2076-2615/11/5/1191)

<https://www.mdpi.com/2076-2615/11/5/1191>

[Rasmussen et al. 2024, Animals: tests and proposed safety test](https://www.mdpi.com/2076-2615/14/1/122)

<https://www.mdpi.com/2076-2615/14/1/122>

[Leibniz-IZW: hedgehogs injured by robot mowers](https://www.izw-berlin.de/en/press-release/new-research-into-hedgehogs-injured-by-robotic-lawn-mowers-discovers-a-significant-but-solvable-animal-welfare-and-conservation-problem.html)

<https://www.izw-berlin.de/en/press-release/new-research-into-hedgehogs-injured-by-robotic-lawn-mowers-discovers-a-significant-but-solvable-animal-welfare-and-conservation-problem.html>

[Hedgehog Street: Near Threatened](https://www.hedgehogstreet.org/near-threatened/)

<https://www.hedgehogstreet.org/near-threatened/>

[Wallonie: rule of 9 April 2026](https://environnement.wallonie.be/actualite/robots-tondeuses-une-regle-wallonne-pour-mieux-proteger-les-herissons)

<https://environnement.wallonie.be/actualite/robots-tondeuses-une-regle-wallonne-pour-mieux-proteger-les-herissons>

[NOS: Nederland volgt nog niet](https://nos.nl/artikel/2609904-egels-in-wallonie-beschermd-tegen-robotmaaiers-nederland-volgt-nog-niet)

<https://nos.nl/artikel/2609904-egels-in-wallonie-beschermd-tegen-robotmaaiers-nederland-volgt-nog-niet>

[Egelbescherming Nederland: winterslaap en bijvoeren](https://www.egelbescherming.nl/winterslaap-en-bijvoeren/)

<https://www.egelbescherming.nl/winterslaap-en-bijvoeren/>

[Egelbescherming Nederland: advice on mowing and bans](https://www.egelbescherming.nl/goed-nieuws-voor-de-egel-bescherming-komt-hoger-op-de-politieke-agenda/)

<https://www.egelbescherming.nl/goed-nieuws-voor-de-egel-bescherming-komt-hoger-op-de-politieke-agenda/>

What already exists

[Husqvarna AI vision for 2026](https://www.husqvarnagroup.com/en/press/husqvarna-group-reveals-ai-vision-technology-robotic-lawnmowers-2026-2397307)

<https://www.husqvarnagroup.com/en/press/husqvarna-group-reveals-ai-vision-technology-robotic-lawnmowers-2026-2397307>

[Navimow hedgehog avoidance claim](https://roboticsandautomationnews.com/2026/05/12/navimow-responds-to-growing-hedgehogs-safety-concerns-with-ai-avoidance-technology/101404/)

<https://roboticsandautomationnews.com/2026/05/12/navimow-responds-to-growing-hedgehogs-safety-concerns-with-ai-avoidance-technology/101404/>

[Retrofit guard that stopped docking, robomaeher.de](https://robomaeher.de)

<https://robomaeher.de/blog/igelschutz-apfelschutz-fuer-maehroboter-rasenroboter/>

Planning

[Soapbox challenge page: dates, "Spots: 100", online kick-off](https://soapbox.nl/challenges/challenge-yHRcBY86HvIM/)

<https://soapbox.nl/challenges/challenge-yHRcBY86HvIM/>

[TU Delft academic calendar 2026-2027 \(PDF\)](https://filelist.tudelft.nl/Studentenportal/Centraal/Mijn%20studie%20_%20ik/Onderwijs/Academische%20jaarindeling/Academic%20Calendar%202026-2027.pdf)

https://filelist.tudelft.nl/Studentenportal/Centraal/Mijn%20studie%20_%20ik/Onderwijs/Academische%20jaarindeling/Academic%20Calendar%202026-2027.pdf

[TU Delft course periods 2026-2027: Q2 starts 16 November](https://www.x.tudelft.nl/en/news/cursusperioden-2026-2027)

<https://www.x.tudelft.nl/en/news/cursusperioden-2026-2027>

[Marktplaats: used Automower 105 listings](https://www.marktplaats.nl/l/tuin-en-terras/robotmaaiers/q/husqvarna+automower+105/)

<https://www.marktplaats.nl/l/tuin-en-terras/robotmaaiers/q/husqvarna+automower+105/>

[Coconut mat on a roll, rubbermatten24.nl](https://rubbermatten24.nl/kokosmat-op-rol-200cm)

<https://rubbermatten24.nl/kokosmat-op-rol-200cm>